

CS336 Assignment 3 (scaling): Scaling Laws

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1 IsoFLOPs Scaling Laws (chinchilla_isoflops)

(a) Optimal Model Size

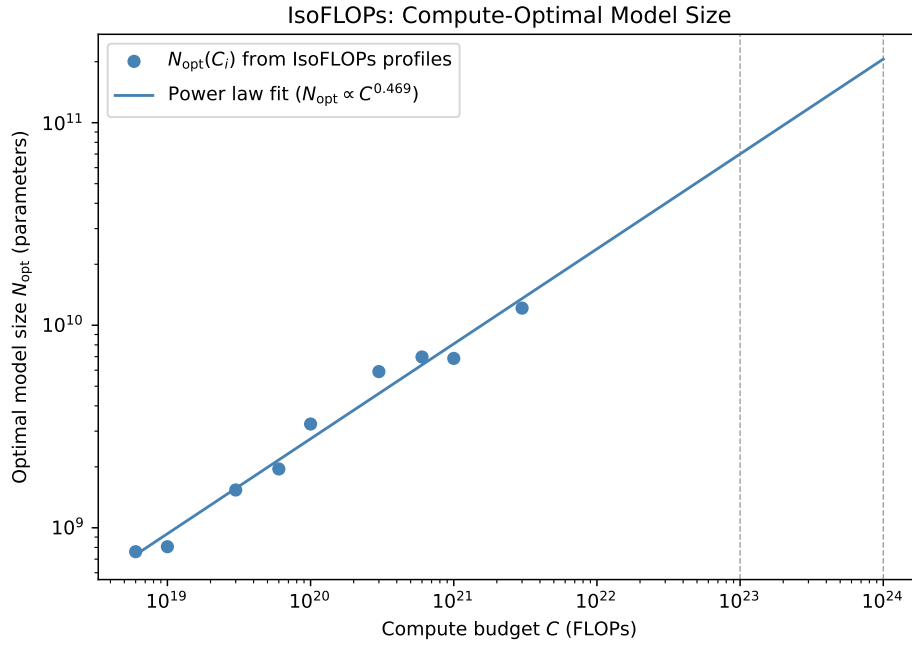


Figure 1: Compute-optimal model size N_{opt} as a function of compute budget C . Data points show $\langle C_i, N_{\text{opt}}(C_i) \rangle$ from the IsoFLOPs profiles; the curve shows the fitted power law $N_{\text{opt}} \propto C^a$.

The predicted compute-optimal model size is 7.0×10^{10} parameters for a budget of 10^{23} FLOPs and 2.1×10^{11} parameters for a budget of 10^{24} FLOPs.

(b) Optimal Dataset Size

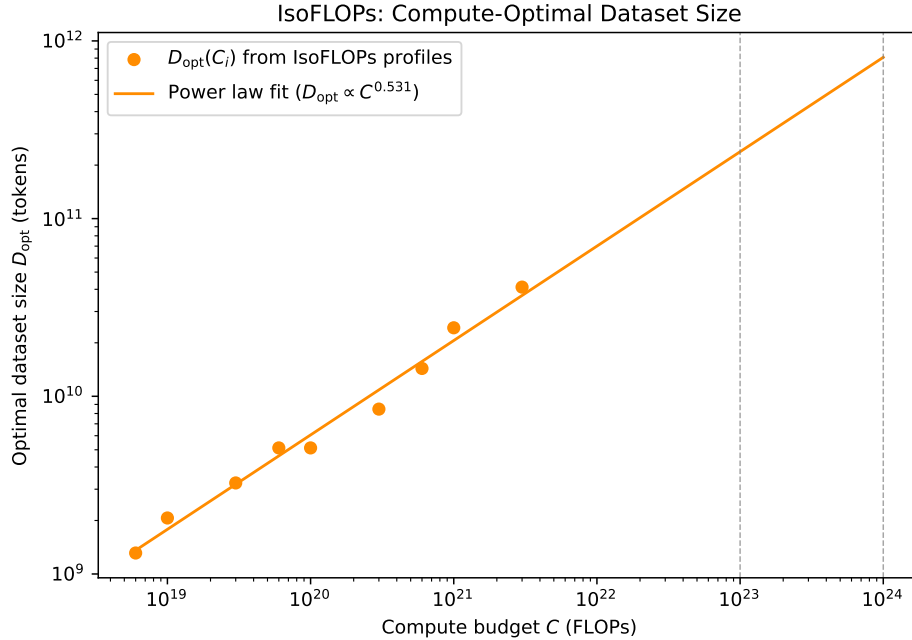


Figure 2: Compute-optimal dataset size D_{opt} as a function of compute budget C . Data points show $\langle C_i, D_{\text{opt}(C_i)} \rangle$ from the IsoFLOPs profiles; the curve shows the fitted power law $D_{\text{opt}} \propto C^b$.

The predicted compute-optimal dataset size is 2.4×10^{11} tokens for a budget of 10^{23} FLOPs and 8.1×10^{11} tokens for a budget of 10^{24} FLOPs.

2 Constructing Scaling Laws (scaling_laws)

2.1 Experimental Design

[PLACEHOLDER: describe how you allocated your 12 B200-hour scaling-law budget — which model sizes, token counts, and hyperparameter settings you queried, and why.]

2.2 Fitting the Scaling Law

[PLACEHOLDER: describe the method used to fit the scaling law (e.g. IsoFLOPs / Chinchilla approach, power-law fit via `scipy.optimize.curve_fit`, etc.) and any design decisions made.]

2.3 Quality of Fit

[PLACEHOLDER: plot showing fitted scaling law against experimental data points]

Figure 3: Fitted scaling law overlaid on experimental validation losses.

[PLACEHOLDER: brief quantitative assessment of how well the scaling law fits the data (e.g. R^2 , residuals).]

2.4 Predicted Optimal Configuration for 48 B200-Hours

[PLACEHOLDER: state your predicted optimal model size N^* and the predicted validation loss for the 48 B200-hour run.]

The predicted compute-optimal number of parameters is $N^* = [\text{PLACEHOLDER}]$, corresponding to a predicted final validation loss of $L^* = [\text{PLACEHOLDER}]$.

2.5 Hyperparameter Choices

[PLACEHOLDER: describe the architecture and optimizer hyperparameters you would use for the predicted optimal model. Justify choices (learning rate, batch size, number of layers, hidden size, etc.), referencing any smaller-scale experiments you ran to inform them. Recall that the non-embedding parameter count is approximated by $12n_{\text{layer}}d_{\text{model}}^2$.]

Hyperparameter	Value
<code>num_hidden_layers</code>	[PLACEHOLDER]
<code>hidden_size</code>	[PLACEHOLDER]
<code>num_attention_heads</code>	[PLACEHOLDER]
<code>head_dim</code>	[PLACEHOLDER]
<code>intermediate_size</code>	[PLACEHOLDER]
<code>total_train_tokens</code>	[PLACEHOLDER]
<code>train_batch_size</code>	[PLACEHOLDER]
Peak learning rate	[PLACEHOLDER]

Table 1: Predicted optimal hyperparameters for the 48 B200-hour run.

References

- [1] J. Kaplan *et al.*, “Scaling Laws for Neural Language Models.” 2020.
- [2] J. Hoffmann *et al.*, “Training Compute-Optimal Large Language Models.” 2022.
- [3] G. Yang *et al.*, “Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer.” 2022.